A7q4 comsol

Report date

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1 Global Definitions

Date	Sep 20, 2025, 12:46:59 AM

GLOBAL SETTINGS

Name	A7q4 comsol.mph
Path	E:\Comsol codes\a7q4_comsol.mph
Version	COMSOL Multiphysics 6.0 (Build: 318)

USED PRODUCTS

COMSOL Multiphysics

COMPUTER INFORMATION

CPU	Intel64 Family 6 Model 142 Stepping 12, 2 cores
Operating system	Windows 10

1.1 PARAMETERS

PARAMETERS 1

Name	Expression	Value	Description
Re	1e3	1000	

2 Component 1

2.1 DEFINITIONS

2.1.1 Coordinate Systems

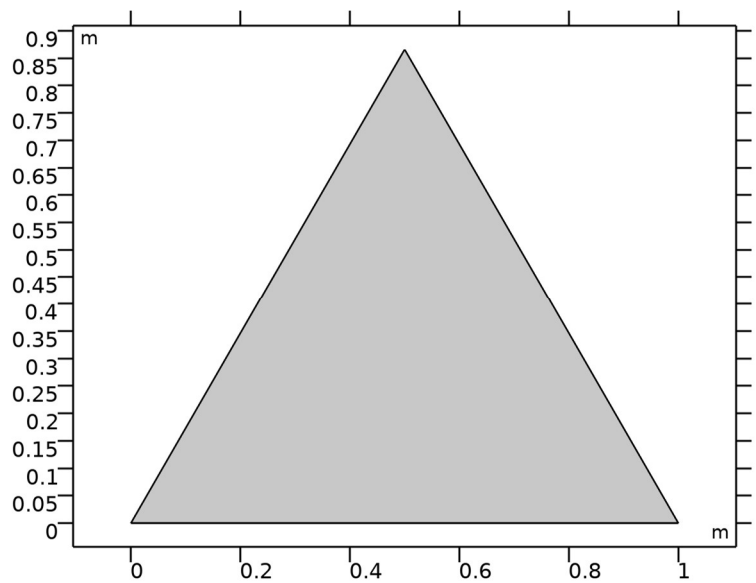
Boundary System 1

Coordinate system type	Boundary system
Tag	sys1

COORDINATE NAMES

First	Second	Third
t1	n	to

2.2 GEOMETRY 1

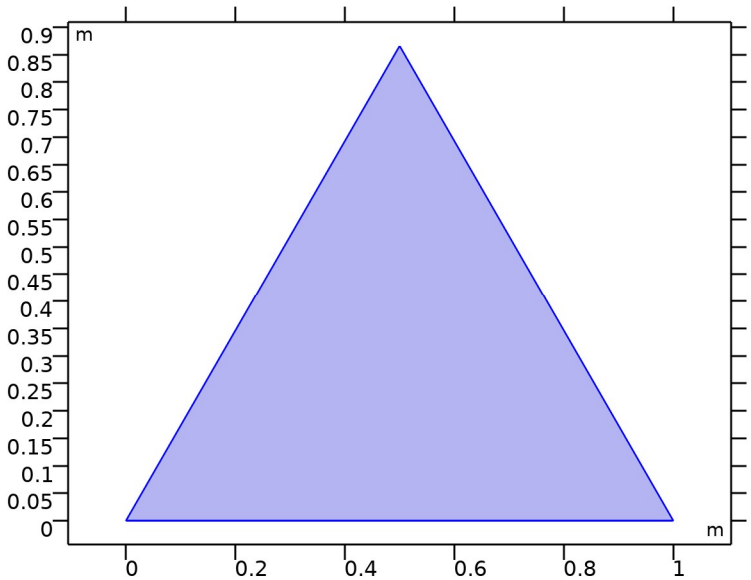


Geometry 1

UNITS

Length unit	m
Angular unit	deg

2.3 LAMINAR FLOW



Laminar Flow

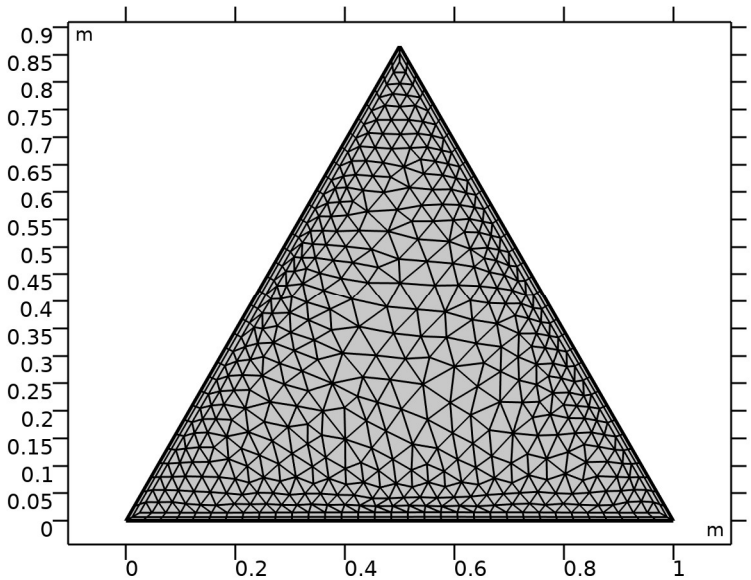
EQUATIONS

$$\rho(\mathbf{u}\cdot\nabla)\mathbf{u} = \nabla\cdot[-p\mathbf{I} + \mathbf{K}] + \mathbf{F}$$
$$\rho\nabla\cdot\mathbf{u} = 0$$

FEATURES

Name	Level
Fluid Properties 1	Domain
Initial Values 1	Domain
Wall 1	Boundary
Pressure Point Constraint 1	Point

2.4 MESH 1



Mesh 1

3 Study 1

COMPUTATION INFORMATION

Computation time	2 s

3.1 STATIONARY

STUDY SETTINGS

Description	Value
Include geometric nonlinearity	Off

PHYSICS AND VARIABLES SELECTION

Physics interface	Discretization
Laminar Flow (spf)	physics

MESH SELECTION

Geometry	Mesh
Geometry 1 (geom1)	mesh1

4 Study 2

COMPUTATION INFORMATION

Computation time	2 s

4.1 PARAMETRIC SWEEP

Parameter name	Parameter value list
Re	1e4 1e5 1e7

STUDY SETTINGS

Description	Value
Sweep type	Specified combinations
Parameter name	Re
Unit	

PARAMETERS

Parameter name	Parameter value list	Parameter unit
Re	1e4 1e5 1e7	

4.2 STATIONARY

STUDY SETTINGS

Description	Value
Include geometric nonlinearity	Off

PHYSICS AND VARIABLES SELECTION

Physics interface	Discretization
Laminar Flow (spf)	physics

MESH SELECTION

Geometry	Mesh
Geometry 1 (geom1)	mesh1

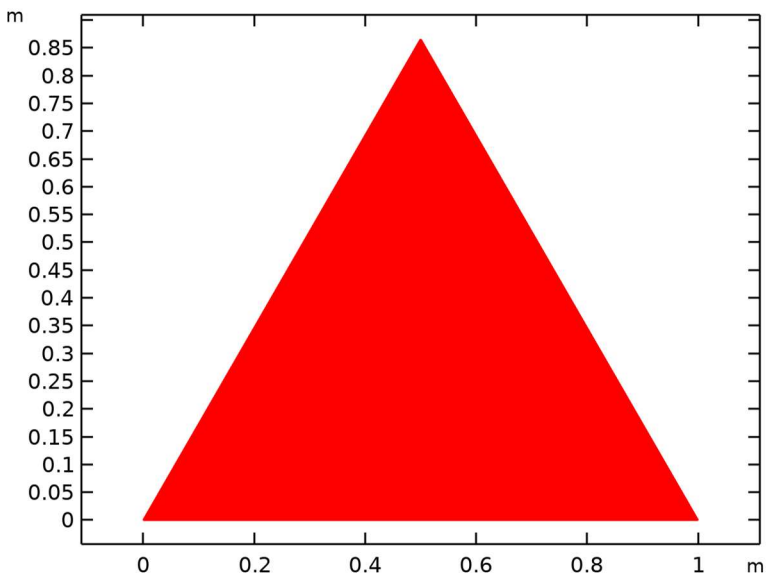
5 Results

5.1 DATASETS

5.1.1 Study 1/Solution 1

SOLUTION

Description	Value
Solution	Solution 1
Component	Component 1 (comp1)

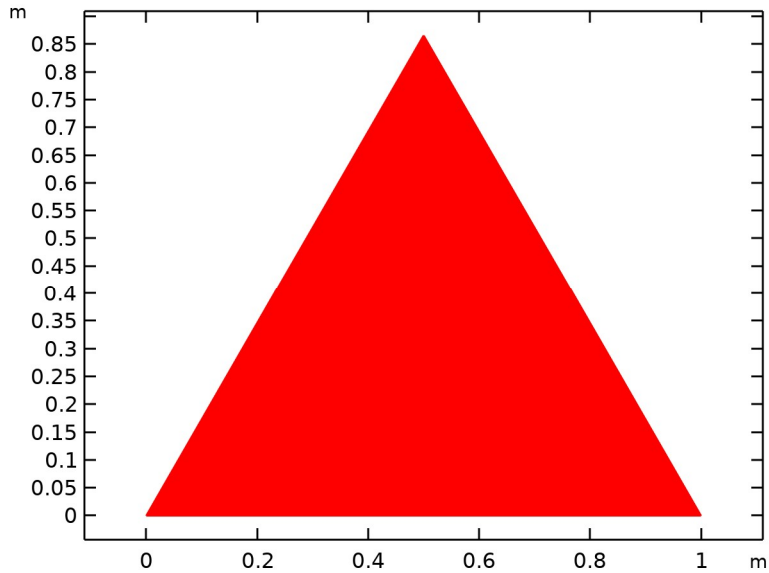


Dataset: Study 1/Solution 1

5.1.2 Study 2/Solution 2

SOLUTION

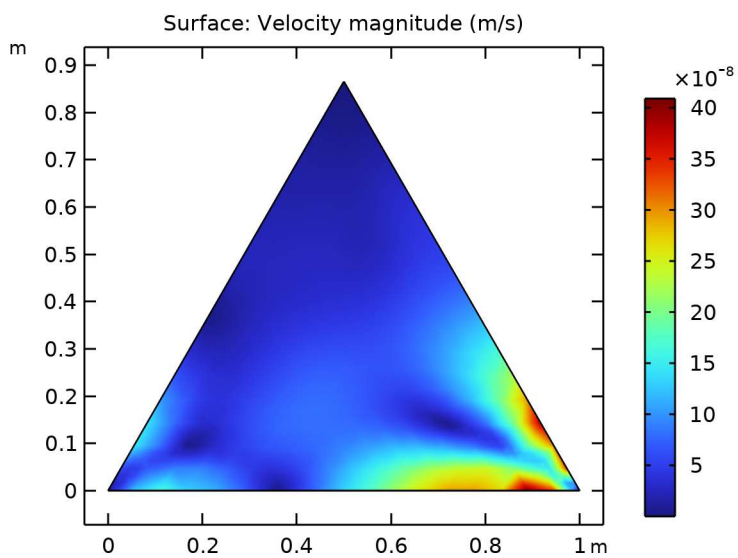
Description	Value
Solution	Solution 2
Component	Component 1 (comp1)



Dataset: Study 2/Solution 2

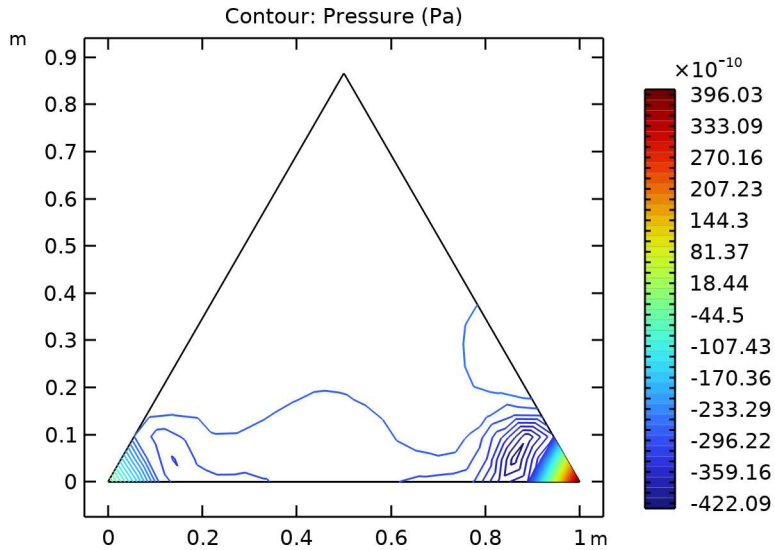
5.2 PLOT GROUPS

5.2.1 Velocity (spf)



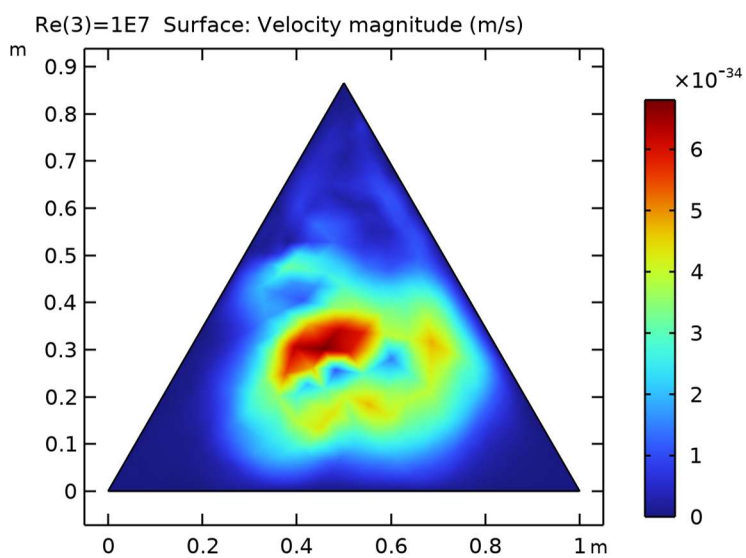
Surface: Velocity magnitude (m/s)

5.2.2 Pressure (spf)



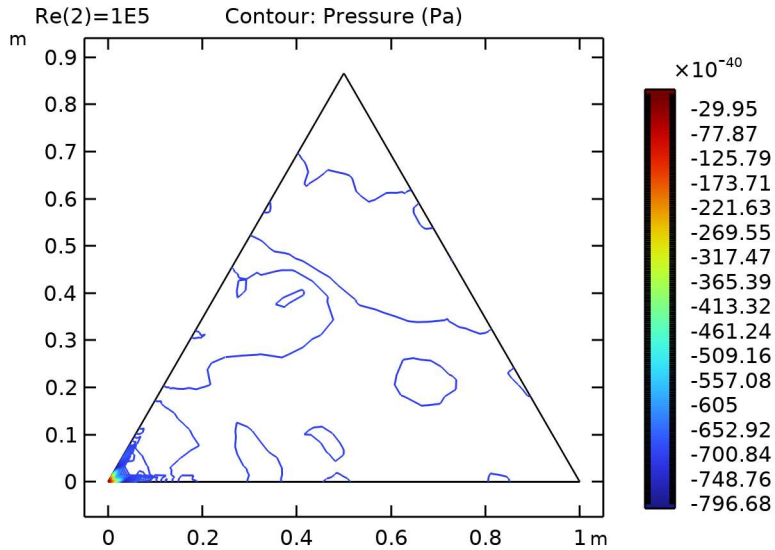
Contour: Pressure (Pa)

5.2.3 Velocity (spf) 1



Surface: Velocity magnitude (m/s)

5.2.4 Pressure (spf) 1



Contour: Pressure (Pa)